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DIPLOMA IN COMPUTER ENGINEERING

A Project Synopsis Report On

COLLEGE EVENT MANAGEMENT SYSTEM

Submitted By:

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1. INTRODUCTION & PROJECT OVERVIEW

The **College Event Management System** is developed as a comprehensive solution designed to automate and simplify the management framework of campus-wide events, academic seminars, sports tournaments, and cultural programs. Organizing multiple student and institutional events manually often places a significant burden on administration staff, resulting in scheduling conflicts, coordination issues, and unorganized data files.

This software platform addresses these core issues by providing a dynamic, centralized environment that keeps organized records of events and user registrations. The web-based framework allows for effortless maintenance of critical event profiles, including specific titles, timeline boundaries, designated locations, and point-of-contact details. Implemented via a full-stack environment consisting of an Angular view layout, a Python Flask processing layer, and a structured MySQL relational engine managed through SQLAlchemy, the system removes paper dependencies and provides instantaneous data availability.

2. PROBLEM STATEMENT

The manual handling of campus operations remains highly inefficient, prone to human error, and time-intensive. Relying on basic hardcopy spreadsheets or paperwork lists poses severe challenges across the operational timeline:

- **Data Loss Risk:** Storing event and registration listings physically creates high exposure to information misplacement or absolute data loss.
- **Logistical Bottlenecks:** Coordinating registration numbers and assigning infrastructure slots manually introduces unexpected overlap conflicts.
- **Scalability Limitations:** As event lists expand over an academic session, tracking structural details, active participants, and assigned coordinators becomes unmanageable.
- **Redundancy & Entry Errors:** Manual data-entry pipelines often lead to duplicate logs, typo faults, and mismatch issues within rosters.

3. PROJECT OBJECTIVES

The structural implementation aims to meet specific architectural benchmarks to optimize campus operations:

- **Centralized Administration Console:** Deliver a single, secure dashboard interface allowing administrators to build, update, review, and delete event items directly.
- **Simplified Registration Mechanics:** Facilitate responsive web portals enabling students to search for activities and register smoothly online.

- **Eliminate Manual Paper Processing:** Transition institutional tracking to completely digital storage files, minimizing human entry issues.
- **Conflict-Free Scheduling:** Automate workspace slot allocations based on calendar dates, time markers, and structural venues.
- **Automated Matrix Reporting:** Empower faculty with real-time tools to compile participation metrics, check entry logs, and extract performance metrics.

4. SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATIONS

The operational environment is constructed with modern software and hardware parameters to ensure stability:

4.1 Requirements Specifications

Category Type	System Requirement Standards
Operating System	Windows 10 / Windows 11 Platform
Programming Languages	HTML, CSS, JavaScript, Python, TypeScript
Frontend Framework	Angular Environment
Backend Microservice	Python Flask Core
Database / ORM Layer	MySQL Server interacting through SQLAlchemy ORM
Integrated Development Tool	Visual Studio Code (VS Code)
Target Application Browser	Google Chrome Platform Engine
Minimum Hardware Processor	Intel Core i3 Architecture or higher

5. CORE FUNCTIONAL MODULES

The platform establishes distinct software components to manage structural activities securely:

1. **User Authentication System:** Secure, encrypted portal logins designed specifically to isolate student profiles from admin consoles.
2. **Event Management Lifecycle (CRUD):** Gives coordinators complete control to construct new event schedules, update rules, and delete outdated entries.
3. **Interactive Registration Wizard:** Provides a simplified enrollment portal with built-in input validation checks.
4. **Dynamic Search Engine:** Includes filter tools allowing users to quickly sort active listings by keyword criteria or dates.
5. **Analytical KPI Dashboard:** Provides visual statistical summaries showing overall user metrics and event density status.

6. DATABASE SCHEMA DESIGN

The structured relational configuration inside the MySQL engine contains two highly optimized schema models:

6.1 Registrations Table Structure

Column Name	Data Type	Constraint Type	Functional Scope
id	INT	Primary Key, Auto Inc	Uniquely identifies each registration record.
event_id	INT / VARCHAR(15)	NOT NULL	Links the transaction map to the designated event.
student_name	VARCHAR(50)	NOT NULL	Stores the full string name of the registering student.
student_email	VARCHAR(50)	Unique, NOT NULL	Stores student's email and prevents duplicate entries.

6.2 Events Table Structure

Column Name	Data Type	Constraint Type	Functional Scope
id	INT	Primary Key, Auto Inc, NOT NULL	Uniquely identifies each event entry.
title	VARCHAR(50)	NOT NULL	Stores the specific title designation of the event.
category	VARCHAR(50)	NOT NULL	Saves the classifications like Technical, Sports, or Cultural.
description	TEXT	NOT NULL	Houses complete rules, timelines, and details.
date	DATE	NOT NULL	Stores the scheduled date of event execution.
time	TIME	NOT NULL	Defines the precise kickoff timeline.
venue	VARCHAR(100)	NOT NULL	Stores the campus infrastructure location or lab.
organizer	VARCHAR(100)	NOT NULL	Identifies the department or committee in charge.
image	VARCHAR(100)	NULL Allowed	Stores the system file path for promotional banners.

7. FUTURE SCOPE & ROAD MAP

The platform layout is designed to allow for easy updates and additions. Future versions aim to introduce native mobile application builds optimized for Android and iOS systems, offering push notifications for sudden scheduling or room changes. Online payment handling frameworks and ticketing gateways will be added to easily coordinate premium programs. Security check-ins at venue entry points will be upgraded with automated QR code validation protocols. Finally, predictive analytical engines will be explored to suggest events to students based on past choices, and cloud migrations will ensure robust system scaling.

8. CONCLUSION

The **College Event Management System** provides an efficient, reliable, and digital alternative to paper-based campus administration. By eliminating manual dependencies, the platform helps prevent data loss, minimizes processing delays, and improves overall accuracy. Connecting a dynamic Angular frontend with an optimized Python Flask server and a structured MySQL database ensures stable calendar

scheduling, swift registrations, and secure database lookups. Ultimately, the project reduces administrative complexity and enhances operational productivity at **Sahakar Maharshi Shankarrao Mohite Patil Institute of Technology and Research, Akluj.**